

Genetic Analysis Report

Report ID:	SEQ-05019F355AC7
Generated:	2025-12-04 19:21:53
Patient Name:	Genetic Risk Analysis Report - HIGH RISK

Analysis Accuracy

93.3%

Patient Information

Analysis performed on file DNA-E7D3A42DE56C. Processing completed using GENO AI v2.0 Elastic Net Logistic Regression model. Model trained on gene expression data with 94.2% cross-validation accuracy. Risk probability calculated: 95.0%

Variant Information

Gene Expression Analysis Complete • Analyzed Genes: 237 • Mean Expression Level: 67.86 • Expression Variance: 93257.16 • High Expression Markers: 54 • Risk Assessment: HIGH RISK • Confidence: 95.0%

Analysis Results

{"report": {"sample_id": "SEQ-05019F355AC7", "laboratory_user_id": "F55055", "generated_at": "2025-12-04T16:21:52.950634Z", "model_name": "geno_enet_pipeline.pkl", "model_version": "2.0", "total_genes_in_model": 237, "genes_analyzed": 237, "risk_level": "HIGH", "risk_score_percent": 97.87, "risk_score_raw": 0.9786992889390222, "risk_threshold": 75.0, "summary_title": "Genetic Risk Analysis Report - HIGH RISK", "summary_text": "The genetic analysis indicates a HIGH addiction risk profile based on the evaluated gene-expression markers. The model analyzed 237 genetic markers and calculated a risk probability of 97.87%. This score exceeds the 75% threshold, indicating elevated risk.", "top_genes": [], "bottom_genes": [], "gene_stats": {"mean_expression": 67.8565, "median_expression": 0.0, "std_expression": 306.0267, "min_expression": 0.0, "max_expression": 3370.0, "genes_above_mean": 26, "genes_below_mean": 211}, "methodology": {"algorithm": "Elastic Net Logistic Regression", "preprocessing": "StandardScaler normalization", "feature_selection": "SelectKBest with f_classif", "training_samples": "Gene expression dataset", "validation": "Cross-validation"}}, {"result": {"sequence_id": "SEQ-05019F355AC7", "laboratory_user_id": "F55055", "risk_level": "HIGH", "score_percent": 97.87, "model_used": "geno_enet_pipeline.pkl", "genes_used": 237, "methodology": {"algorithm": "Elastic Net Logistic Regression", "preprocessing": "StandardScaler normalization", "feature_selection": "SelectKBest with f_classif", "training_samples": "Gene expression dataset", "validation": "Cross-validation"}}

This report was generated automatically from genetic data analysis. For questions or concerns, please contact your healthcare provider.